

TASK-1 REPORT

Foundations of Ethical Hacking & Penetration Testing

Student Details

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- **Course:** B.Tech CSE (Cyber Security)
- **Task:** Task-1
- **Internship:** Ethical Hacking & Penetration Testing

1. Objective

The objective of this task is to build a strong foundation in ethical hacking and penetration testing by understanding core concepts, setting up a virtual lab environment, and performing basic reconnaissance and scanning.

2. Ethical Hacking Concepts

Ethical hacking involves testing systems for vulnerabilities with proper authorization. It helps organizations improve security.

Key Points:

- Ethical hacking is legal with permission
- Illegal hacking is unauthorized
- Responsible disclosure is important

3. Penetration Testing Methodology

The penetration testing process includes the following phases:

- Reconnaissance – Information gathering
- Scanning – Identifying open ports/services
- Exploitation – Finding vulnerabilities
- Reporting – Documenting results

4. Lab Setup

A virtual lab environment was created using VMware Workstation.

Tools Used:

- Kali Linux (Attacker Machine)
- Metasploitable2 (Target Machine)
- VMware Workstation

Network Configuration:

- Network Type: Host-only (VMnet1)
- IP Range: 192.168.199.xxx

5. Linux Commands Practiced

The following basic Linux commands were executed:

1. pwd
2. ls
3. cd
4. mkdir
5. cp
6. mv
7. rm
8. chmod
9. sudo apt update

These commands help in navigation and file management.

6. Networking Fundamentals

Basic networking concepts learned include:

- IP Address
- Ports (TCP/UDP)
- DNS
- HTTP vs HTTPS

Commands Used: ip a , ping 192.168.199.129

7. Connectivity Testing

The connection between Kali Linux and Metasploitable was verified using ping.

Command:

```
ping 192.168.199.129
```

Result: Successful communication was established with zero packet loss.

8. Nmap Scanning

Nmap tool was used to scan the target system.

Command:

```
nmap -sS -sV 192.168.199.129
```

Observation: The scan helps in identifying open ports and services running on the target system.

9. Challenges Faced

- Network mismatch (NAT vs Host-only)
- Services not running in target machine
- Initial scans showing all ports closed

Solutions:

- Configured Host-only network
- Restarted services in Metasploitable
- Verified connectivity using ping

10. Conclusion

This task provided a clear understanding of ethical hacking concepts and practical implementation. I learned to configure a lab environment and perform basic network scanning. It enhanced my skills in Linux, networking, and problem-solving. This foundation will help me in advanced penetration testing tasks.

11. Screenshots / Evidence

Figure 1: Lab Setup

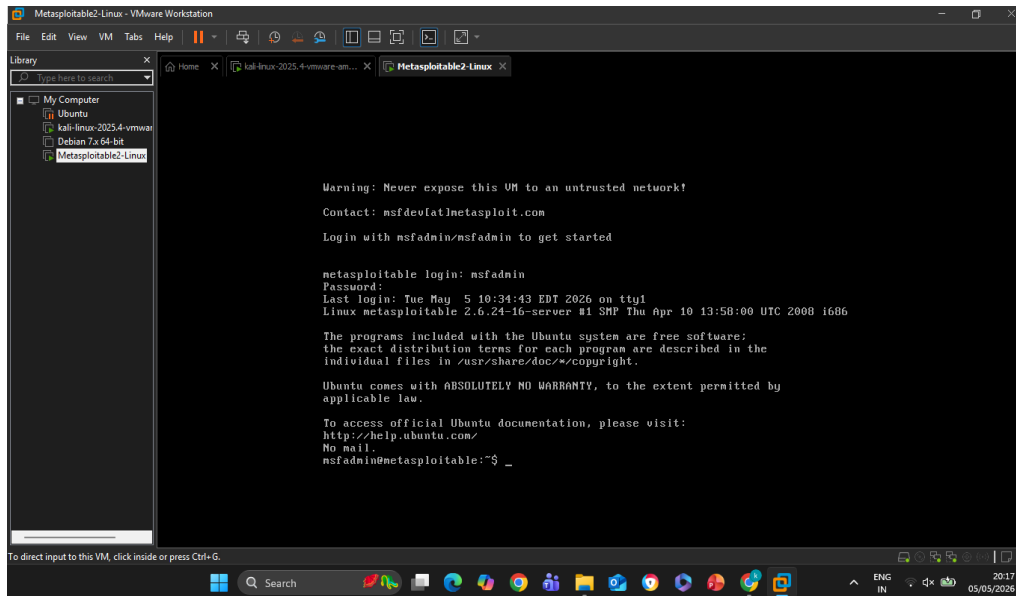


Figure 2: Kali Linux IP Address

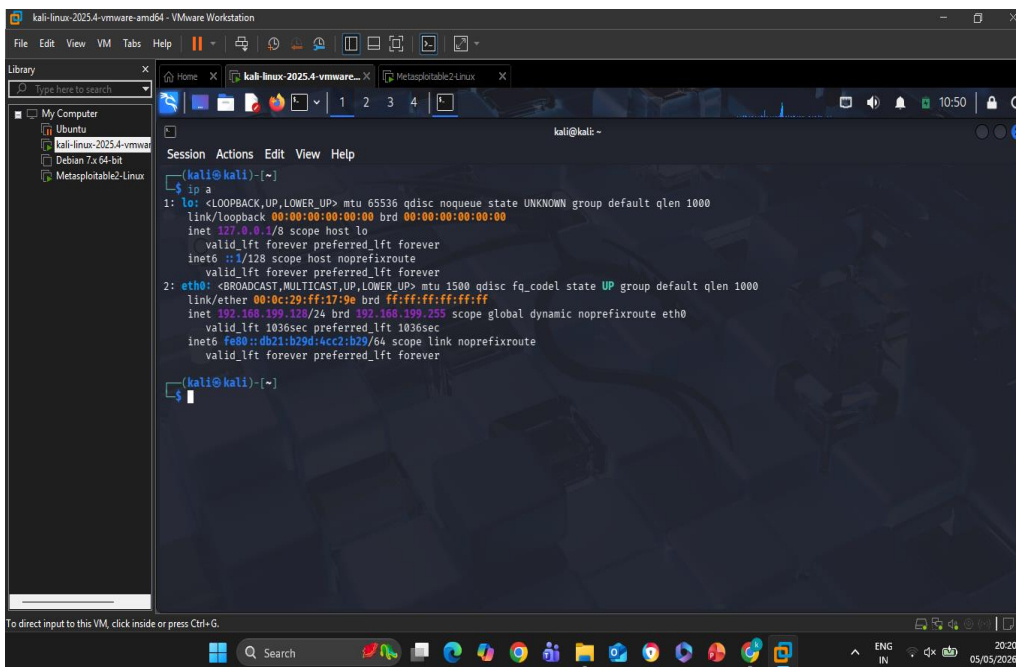
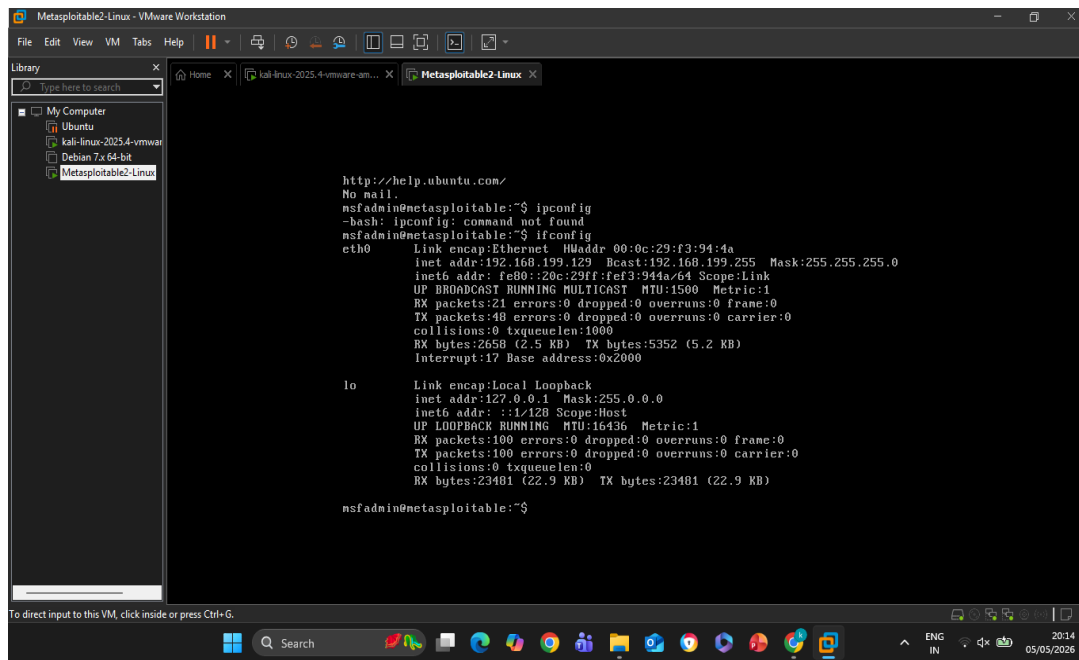


Figure 3: Metasploitable IP Address

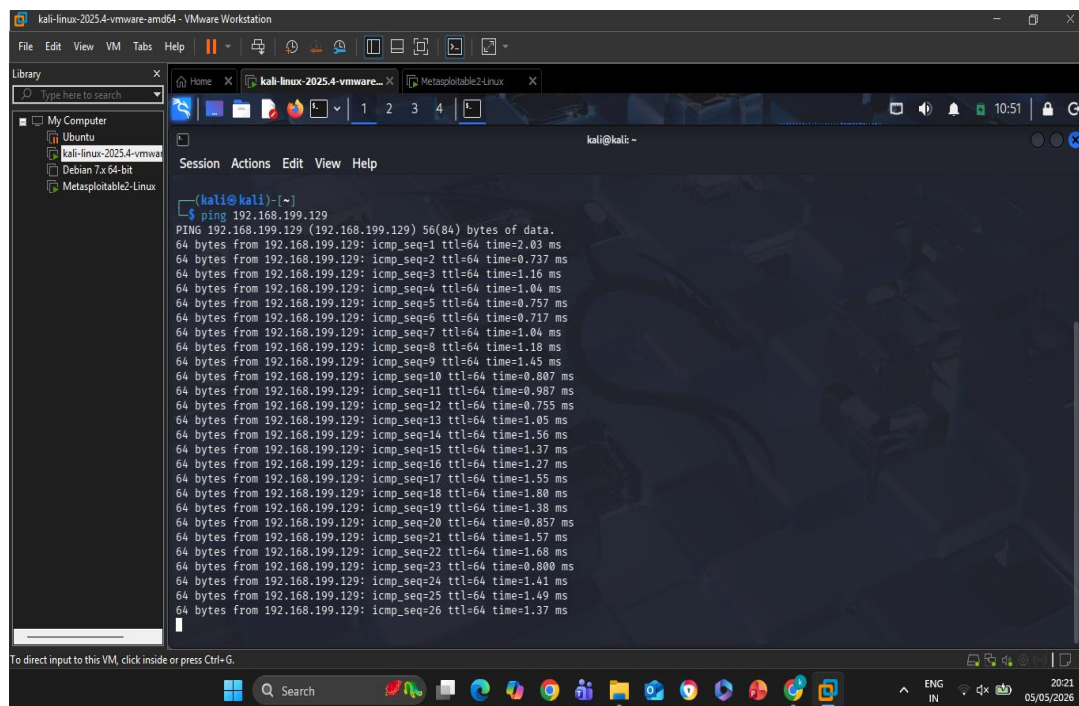


```
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
-bash: ifconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:f3:94:4a
          inet addr:192.168.199.129  Bcast:192.168.199.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe93:944a/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:21 errors:0 dropped:0 overruns:0 frame:0
          TX packets:48 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:2658 (2.5 KB)  TX bytes:5352 (5.2 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:100 errors:0 dropped:0 overruns:0 frame:0
          TX packets:100 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:23481 (22.9 KB)  TX bytes:23481 (22.9 KB)

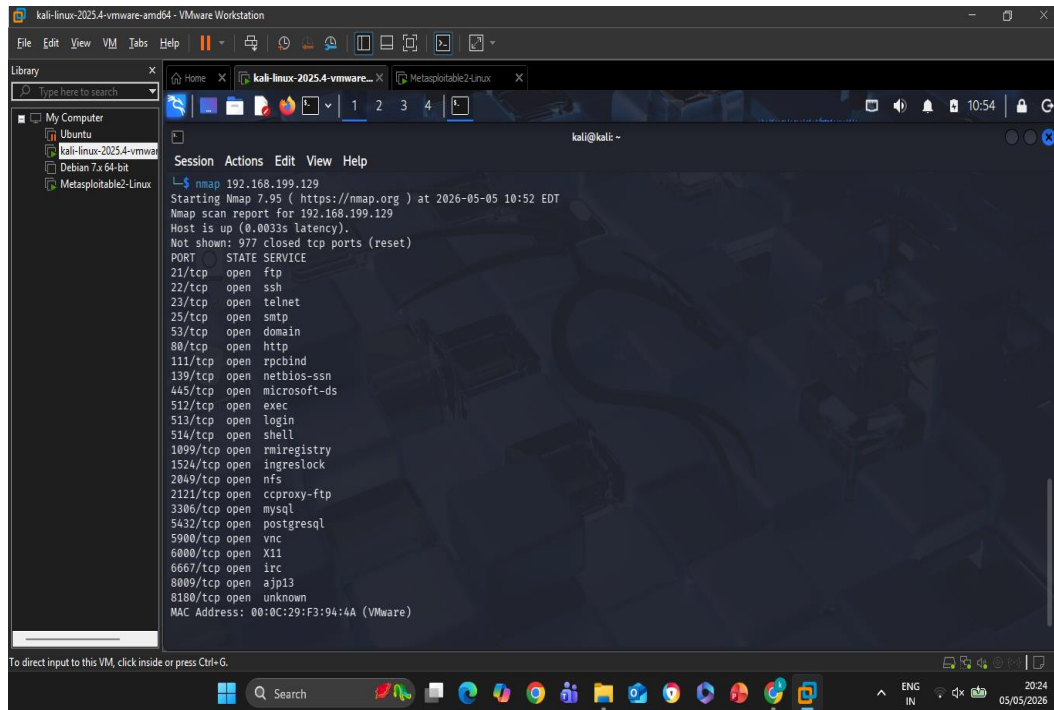
msfadmin@metasploitable:~$
```

Figure 4: Ping Result

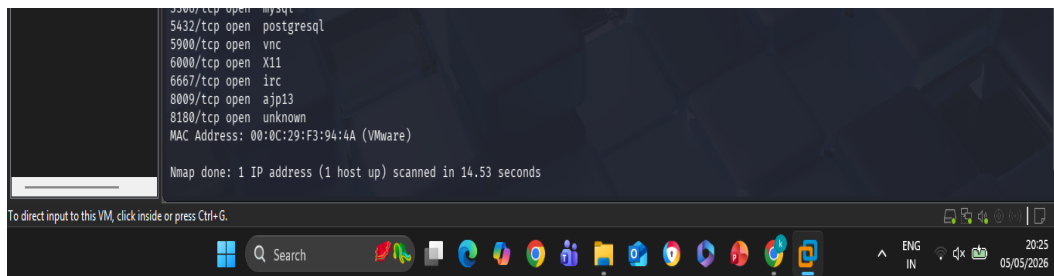


```
kali@kali:~$ ping 192.168.199.129
PING 192.168.199.129 (192.168.199.129) 56(84) bytes of data:
64 bytes from 192.168.199.129: icmp_seq=1 ttl=64 time=2.03 ms
64 bytes from 192.168.199.129: icmp_seq=2 ttl=64 time=0.737 ms
64 bytes from 192.168.199.129: icmp_seq=3 ttl=64 time=1.16 ms
64 bytes from 192.168.199.129: icmp_seq=4 ttl=64 time=1.04 ms
64 bytes from 192.168.199.129: icmp_seq=5 ttl=64 time=0.757 ms
64 bytes from 192.168.199.129: icmp_seq=6 ttl=64 time=0.717 ms
64 bytes from 192.168.199.129: icmp_seq=7 ttl=64 time=1.04 ms
64 bytes from 192.168.199.129: icmp_seq=8 ttl=64 time=1.18 ms
64 bytes from 192.168.199.129: icmp_seq=9 ttl=64 time=1.45 ms
64 bytes from 192.168.199.129: icmp_seq=10 ttl=64 time=0.807 ms
64 bytes from 192.168.199.129: icmp_seq=11 ttl=64 time=0.987 ms
64 bytes from 192.168.199.129: icmp_seq=12 ttl=64 time=0.755 ms
64 bytes from 192.168.199.129: icmp_seq=13 ttl=64 time=1.05 ms
64 bytes from 192.168.199.129: icmp_seq=14 ttl=64 time=1.56 ms
64 bytes from 192.168.199.129: icmp_seq=15 ttl=64 time=1.37 ms
64 bytes from 192.168.199.129: icmp_seq=16 ttl=64 time=1.27 ms
64 bytes from 192.168.199.129: icmp_seq=17 ttl=64 time=1.55 ms
64 bytes from 192.168.199.129: icmp_seq=18 ttl=64 time=1.80 ms
64 bytes from 192.168.199.129: icmp_seq=19 ttl=64 time=1.38 ms
64 bytes from 192.168.199.129: icmp_seq=20 ttl=64 time=0.857 ms
64 bytes from 192.168.199.129: icmp_seq=21 ttl=64 time=1.57 ms
64 bytes from 192.168.199.129: icmp_seq=22 ttl=64 time=1.68 ms
64 bytes from 192.168.199.129: icmp_seq=23 ttl=64 time=0.800 ms
64 bytes from 192.168.199.129: icmp_seq=24 ttl=64 time=1.41 ms
64 bytes from 192.168.199.129: icmp_seq=25 ttl=64 time=1.49 ms
64 bytes from 192.168.199.129: icmp_seq=26 ttl=64 time=1.37 ms
```

Figure 5: Nmap Scan Result



```
kali@kali: ~  
└─$ nmap 192.168.199.129  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-05 10:52 EDT  
Nmap scan report for 192.168.199.129  
Host is up (0.0033s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  mircregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 00:0C:29:F3:94:4A (VMware)
```



```
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 00:0C:29:F3:94:4A (VMware)  
  
Nmap done: 1 IP address (1 host up) scanned in 14.53 seconds
```

End of Report